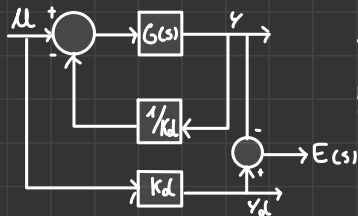


COMPORTAMENTO A REGIME



$$y = y_d$$

$$\lim_{t \rightarrow \infty} e(t) = 0$$

CI SPOSTIAMO SUL PIANO S

$$E(s) = -W(s)U(s) + K_d U(s)$$

$$W(s) = \frac{G(s)}{1 + \frac{1}{K_d} \cdot G(s)}$$

$$= \left(K_d - \frac{G(s)}{1 + \frac{1}{K_d} \cdot G(s)} \right) U(s) = \left(K_d - \frac{K_d G(s)}{K_d + G(s)} \right) U(s) = \left(\frac{K_d^2}{K_d + G(s)} \right) U(s)$$

APPLICHIAMO IL TEOREMA DEL VALORE FINALE

$$\lim_{s \rightarrow 0} s E(s) U(s) = \lim_{s \rightarrow 0} s \cdot \left(\frac{K_d^2}{K_d + G(s)} \right) \cdot \frac{1}{s^{k+1}}$$

$$u(t) = \frac{t^k}{k!} \xrightarrow{L} U(s) = \frac{1}{s^{k+1}}$$

$$= \lim_{s \rightarrow 0} \frac{K_d^2}{K_d + \frac{G(s)}{s^h}} \cdot \frac{1}{s^k} = \lim_{s \rightarrow 0} \frac{K_d^2}{s^h K_d + G(s)} \cdot \frac{1}{s^{h+k}}$$

$$G(s) = C(s)P(s) \rightarrow G(s) = \frac{G'(s)}{s^h}$$

$$G'(0) = K_0$$

$$h > k \quad e(\infty) = 0$$

$$h < k \quad e(\infty) = \infty$$

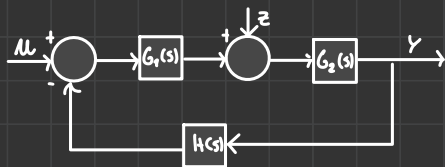
$$h = k \quad h = 0 \quad \frac{K_d^2}{K_d + K_0}$$

$$h \geq 1 \quad \frac{K_d^2}{K_0}$$

TABELLINA

	$\delta_{-1}(t)$	$\delta_{-2}(t)$	$\delta_{-3}(t)$
$h \setminus k$	$\frac{K_d^2}{K_d + K_0}$	$\frac{K_d^2}{K_0}$	$\frac{K_d^2}{K_0}$
0	$\frac{K_d^2}{K_d + K_0}$	∞	∞
1	0	$\frac{K_d^2}{K_0}$	∞
2	0	0	$\frac{K_d^2}{K_0}$

REIEZIONE DEI DISTURBI



$$W_z(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)}$$

$$Z(s) = \frac{1}{s^{k+1}}$$

APPLICHIAMO IL TEOREMA DEL VALORE FINALE

$$\lim_{s \rightarrow 0} s W_z(s) Z(s) = \lim_{s \rightarrow 0} \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} \frac{1}{s^{k+1}} = \lim_{s \rightarrow 0}$$

$$G_1(s) = \frac{G'_1(s)}{s^h}$$

$$\frac{G_2(s)}{1 + \frac{G'_1(s)}{s^h} G_2(s) H(s)} \frac{1}{s^k} = \lim_{s \rightarrow 0} \frac{G_2(s)}{s^h + G'_1(s) G_2(s) H(s)} \cdot \frac{1}{s^{h+k}}$$

$$G'_1(0) = k_{G_1} \quad G_2(0) = k_{G_2}$$

$$h > k \quad e_z(\infty) = 0$$

$$h < k \quad e_z(\infty) = \infty$$

$$h = k \rightarrow e_z(\infty) = \frac{k_{G_2}}{k_{G_1} k_{G_2} \cdot \frac{1}{k_d}}$$

Diagrammi di Bode

$$G(s) = K_G s^h \frac{\left(\frac{s^2 + 2\zeta}{\omega_n^2} + \frac{2\zeta}{\omega_n^2} s + 1\right) (s\tau + 1)}{\left(\frac{s^2 + 2\zeta}{\omega_n^2} + \frac{2\zeta}{\omega_n^2} s + 1\right) (s\tau + 1)}$$

$$s = j\omega$$

GUADAGNO COSTANTE

$$|K_G|_{dB} = 20 \log_{10} |K_G|$$

$$\angle K_G \rightarrow \begin{matrix} K_G > 0 & 0^\circ \\ K_G < 0 & -180^\circ \end{matrix}$$

ZERI E POLI NEW ORIGIN

$$s^h \rightarrow (j\omega)^h$$

$$|(j\omega)^h|_{dB} = |(j\omega)^h| = 20h \log_{10} \omega = 20h\lambda$$

$$\angle (j\omega)^h = 90h^\circ \quad \lambda = \log_{10} \omega \leftrightarrow \omega = 10^\lambda$$

ZERI E POLI REALI

$$(s\tau + 1) \xrightarrow{s=j\omega} (j\omega\tau + 1)$$

$$|j\omega\tau + 1|_{dB} = \sqrt{(j\omega\tau)^2 + 1} = 10 \log_{10} ((\omega\tau)^2 + 1)$$

$$\omega\tau \gg 1 \quad 20 \log_{10} (\omega\tau) = 20 \log_{10} (\omega) + 20 \log_{10} (\tau) \sim 20 \log_{10} (\omega) = 20\lambda$$

$$\omega\tau \ll 1 \quad 10 \log_{10} (1) = 0 \text{ dB}$$

$$\omega\tau \sim 1$$

1/\tau PUNTO DI ROTAZIONE

FASE

$$\omega\tau \ll 1 \quad \angle(j\omega\tau + 1) \sim \angle(1) = 0^\circ$$

$$\omega\tau \gg 1 \quad \angle(j\omega\tau + 1) \sim \angle(j\omega\tau) = 90^\circ$$

$$\omega\tau = 1 \quad \angle(j\omega\tau + 1) \sim \angle(j+1) = 45^\circ$$

ZERI E POLI COMPRESSI E COMUNITI

$$\left(\frac{s^2}{\omega_m^2} + \frac{2\zeta}{\omega_m} s + 1 \right) \xrightarrow{s = j\omega} \left(\frac{2\zeta\omega}{\omega_m} j + 1 - \frac{\omega^2}{\omega_m^2} \right)$$

MODULO

$$\left| \left(\frac{2\zeta\omega}{\omega_m} j + 1 - \frac{\omega^2}{\omega_m^2} \right) \right|_{dB} = \sqrt{\left(\frac{2\zeta\omega}{\omega_m} \right)^2 + \left(1 - \frac{\omega^2}{\omega_m^2} \right)^2} = 20 \log_{10} \left(\left(\frac{2\zeta\omega}{\omega_m} \right)^2 + \left(1 - \frac{\omega^2}{\omega_m^2} \right)^2 \right)$$

$\omega \gg \omega_m$

$$40 \log_{10} \left(\frac{\omega}{\omega_m} \right) = 40 \log_{10}(\omega) - 40 \log_{10}(\omega_m) \sim 40 \log_{10}(\omega) = 40\lambda$$

$\omega \ll \omega_m$

$$|1| = 0 \text{ dB}$$

$\omega = \omega_m$

$$20 \log_{10} \left(\frac{2\zeta\omega}{\omega_m} \right) = 20 \log_{10}(2\zeta) \quad 0 < \zeta < 1$$

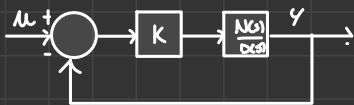
FASE

$$\omega \gg \omega_m \quad \angle -\frac{\omega^2}{\omega_m^2} = 180^\circ$$

$$\omega \ll \omega_m \quad \angle(1) = 0^\circ$$

$$\omega = \omega_m \quad \sim \angle(2\zeta s) = 90^\circ$$

SISTEMI A FEED BACK UNITARIA

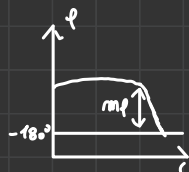
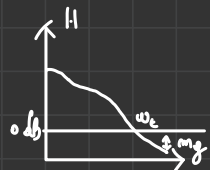


$$W(s) = \frac{K \cdot \frac{N(s)}{D(s)}}{1 + K \cdot \frac{N(s)}{D(s)}} = \frac{K \cdot N(s)}{D(s) + K \cdot N(s)}$$

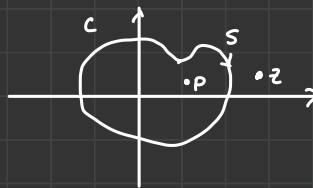
$$K \rightarrow 0$$

$$K \rightarrow \infty$$

MARGINI DI STABILITÀ

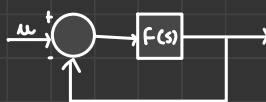


CRITERIO DI NYQUIST



$$P(s) = K \frac{(s-z)}{(s-p)}$$

$$= m^{\circ} \text{ZER}_L(P) - m^{\circ} \text{POL}_L(P)$$



$$W(s) = \frac{F(s)}{1 + F(s)}$$

$$F(s) = \frac{N(s)}{D(s)}$$

$$= \frac{\frac{N(s)}{D(s)}}{1 + \frac{N(s)}{D(s)}} = \frac{N(s)}{D(s) + N(s)}$$

$$1 + F(s) = \frac{D(s) + N(s)}{D(s)}$$

CURVA
PERCORSO
NYQUIST

CRITERIO DI NYQUIST

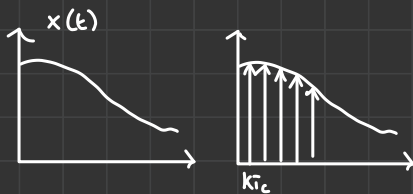
$$h_{(1+F(s)),0} = -m^{\circ} \text{POL}_{\text{DAP}} F(s)$$

$$h_{(F),-1} = -m^{\circ} \text{POL}_{\text{DAP}} F(s)$$

$$h_{(1+F(s)),0} = m^{\circ} \text{ZER}_{\text{DAP}} (1 + F(s)) - m^{\circ} \text{POL}_{\text{DAP}} (1 + F(s))$$

$$= m^{\circ} \text{POL}_{\text{DAP}} W(s) - m^{\circ} \text{POL}_{\text{DAP}} F(s)$$

TEOREMA DEL CAMPIONAMENTO



$$x_c(t) = \sum_{k=-\infty}^{\infty} x(t) \delta(t - kT_c)$$

$$f_c = 1/T_c$$

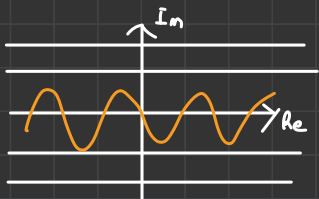
$$X_c(s) = \sum_{k=-\infty}^{\infty} x_k e^{-skT_c}$$

$$\omega_c = 2\pi f_c$$

$$s = \sigma + j\omega$$

$$T_c = \frac{2\pi}{\omega_c}$$

$$X_c(s) = \sum_{k=-\infty}^{\infty} x_k e^{-\sigma k \frac{2\pi}{\omega_c}} e^{-j\omega k \frac{2\pi}{\omega_c}}$$

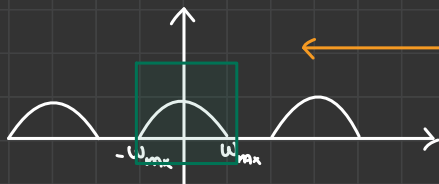


$$f(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} [a_k \cos(k\omega_c t) + b_k \sin(k\omega_c t)]$$

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{-j k \omega_c t} \rightarrow c_k = \frac{1}{T_c} \int_{-\frac{T_c}{2}}^{\frac{T_c}{2}} f(t) e^{-j k \omega_c t} dt = \frac{1}{T_c}$$

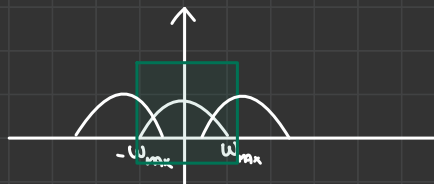
$$x_{c, periodica}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_c} e^{j k \omega_c t}$$

$$x_c(t) = x(t) \cdot \sum_{k=-\infty}^{\infty} \frac{1}{T_c} e^{j k \omega_c t} = \frac{1}{T_c} \sum_{k=-\infty}^{\infty} x(t) e^{j k \omega_c t} \xrightarrow{\mathcal{L}} \frac{1}{T_c} \sum_{k=-\infty}^{\infty} X(s - j k \omega_c) \quad s = j\omega$$



$$\omega_c > 2\omega_n$$

$$\omega_c < 2\omega_n$$



ORGANO DI TERZA DI ORDINE 0

$$x_{\text{SAMPINNO}}(t) = \sum_{k=0}^{\infty} \left[x(t) \delta(t - kT_e) - x(t) \delta(t - (k+1)T_e) \right]$$

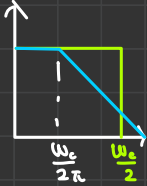
$$X_{\text{SAMPINNO}}(s) = \sum_{k=0}^{\infty} \left[x_k \frac{1}{s} e^{-skT_e} - x_k \frac{1}{s} e^{-s(k+1)T_e} \right] = \left(\sum_{k=0}^{\infty} x_k e^{-skT_e} \right) \left(\frac{1 - e^{-sT_e}}{s} \right)$$

$$T(s) = \left(\frac{1 - e^{-sT_c}}{s} \right)$$

$$= \frac{1 - 1 + sT_c}{s} = \frac{sT_c}{1 + sT_c} = \frac{T_c}{1 + sT_c}$$

$$C^x = 1+x$$

$$e^{sT_c} = 1 + sT_c$$



LINEARIZZAZIONE INTORNO A UN PUNTO D'EQUILIBRIO

$$y = \sin(x) = \sin(x_0) + \frac{d\sin(x_0)}{dx_0} (x - x_0) + \dots$$



$$M d^2 \ddot{\theta} = -Mg d \sin(\theta) - D \dot{\theta} + \tau$$

$$\underline{M d^2 \ddot{\theta} + Mg d \sin(\theta) + D \dot{\theta} = \tau} \quad \theta_0$$

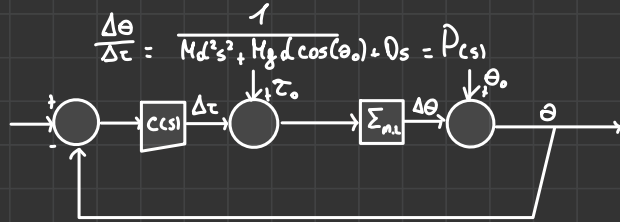
Forza

$$\theta = \theta_0 + \Delta \theta \quad \dot{\theta} = \Delta \dot{\theta} \quad \ddot{\theta} = \Delta \ddot{\theta} \quad \sin(\theta) = \sin(\theta_0) + \cos(\theta_0) \Delta \theta$$

$$M d^2 \Delta \ddot{\theta} + Mg d (\sin(\theta_0) + \cos(\theta_0) \Delta \theta) + D \Delta \dot{\theta} = \tau$$

$$Mg d (\sin(\theta_0)) = \tau_0 \quad \tau = \tau_0 + \Delta \tau$$

$$M_d^2 \Delta \ddot{\theta} + M_g d (\sin(\theta_0) + \cos(\theta_0) \Delta \theta) + D \Delta \dot{\theta} = \tau_0 + \Delta \tau$$



SCelta DEL TEMPO DI CAMPIONAMENTO

$$\omega_c > 2 \omega_{max}$$

$$\omega_c = 2\pi \frac{1}{T_c}$$

$$2\pi \frac{1}{T_c} > 2 \omega_{max}$$

$$\frac{1}{T_c} > \frac{\omega_{max}}{\pi} \rightarrow T_c > \frac{\pi}{\omega_{max}}$$



TRASFORMATA Z

$$y_{k+1} = 3y_k + u_k$$

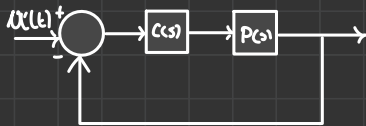
$$Z\{y_k\} = Y_k(z) = \sum_{k=0}^{\infty} y_k z^{-k} = \frac{1}{1 - y_k z^{-k}}$$

$$kT_c$$

$$X_c(s) = \sum_{k=0}^{\infty} x_k e^{-skT_c}$$

$$z = e^{sT_c}$$

$$X_c(z) = \sum_{k=0}^{\infty} x_k z^{-k}$$



$$C(s) = \frac{1}{s-p} \longrightarrow c(t) = e^{pt}$$

$$C(s) = \frac{h_1}{s-p_1} + \frac{h_2}{s-p_2}$$

$$c_c(t) = \sum_{k=0}^{\infty} e^{pkT_c} \delta(t - kT_c)$$

$$C(z) = \frac{h_1}{z - e^{pT_c}} + \dots$$

$$C(z) = \sum_{k=0}^{\infty} e^{pkT_c} z^{-k} = \sum_{k=0}^{\infty} (e^{pT_c} z^{-1})^k = \frac{1}{1 - e^{pT_c} z^{-1}} = \frac{z}{z - e^{pT_c}}$$

$$|e^{pT_c} z^{-1}| < 1$$

$$|z| > e^{pT_c}$$

$$K_{cs} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s-p} \cdot \frac{1}{s} = -\frac{1}{p}$$

$$K_{cz} = \lim_{z \rightarrow 1} (1 - z^{-1}) \frac{z}{z - e^{pT_c}} \cdot \frac{1}{1 - z^{-1}} = \frac{1}{1 - e^{pT_c}}$$

MAPPING DAL PIANO S AL PIANO Z

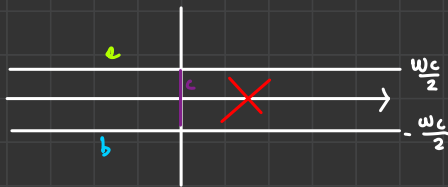
$$X_c(s) = \sum_{k=0}^B x_k e^{-skT_e}$$

$$X_c(z) = \sum_{k=0}^B x_k z^{-k}$$

$$z = e^{sT_e} = e^{\sigma \frac{2\pi}{\omega_c}} e^{j\omega \frac{2\pi}{\omega_c}} = e^{j\omega \frac{2\pi}{\omega_c}}$$

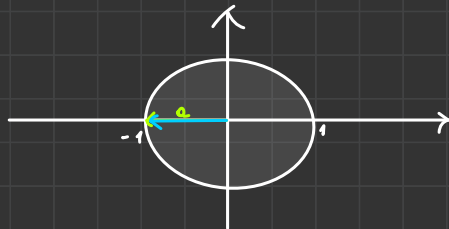
$$e^{j(\frac{\omega_c}{2}) \frac{2\pi}{\omega_c}} = e^{j\pi}$$

$$e^{j(-\frac{\omega_c}{2}) \frac{2\pi}{\omega_c}} = e^{-j\pi}$$



$\text{Re}\{s\} < 0$

$\text{Re}\{s\} = 0$

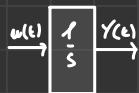
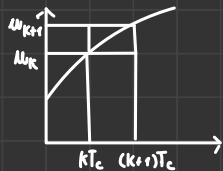


$$\text{Im}\{s\} = 0 \quad z = e^{j\omega \frac{2\pi}{\omega_c}} = 1$$

$$|z| = \sqrt{\sigma^2 + \omega^2}$$

$$|z| \geq 1$$

T TRASFORMAZIONI APPROSSIMATE DAL PIANO S AL PIANO Z



1. DIRETTO

$$y_{k+1} = y_k + \mu_k T_c$$

$$Y(z)(1 - z^{-1}) = T_c U(z) z^{-1}$$

$$\frac{Y(z)}{U(z)} = \frac{T_c z^{-1}}{1 - z^{-1}} \approx \frac{1}{s} \rightarrow s = \frac{1 - z^{-1}}{T_c z^{-1}}$$

2. ECCESSO

$$y_{k+1} = y_k + \mu_{k+1} T_c$$

$$Y(z)(1 - z^{-1}) = T_c U(z)$$

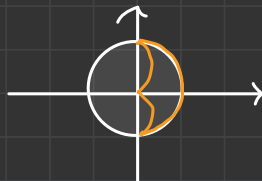
$$\frac{Y(z)}{U(z)} = \frac{T_c}{1 - z^{-1}} \approx \frac{1}{s} \rightarrow s = \frac{1 - z^{-1}}{T_c}$$

3. FORMA TRAPEZOIDALE

$$y_{k+1} = y_k + (\mu_k + \mu_{k+1}) \frac{T_c}{2}$$

$$Y(z)(1 - z^{-1}) = \frac{T_c}{2} U(z) (1 + z^{-1})$$

$$\frac{Y(z)}{U(z)} = \frac{T_c}{2} \frac{1 + z^{-1}}{1 - z^{-1}} \rightarrow s = \frac{2}{T_c} \frac{1 - z^{-1}}{1 + z^{-1}}$$



LINEARIZZAZIONE INTORNO A UN PUNTO D'EQUILIBRIO

$$y = \sin(x) = \sin(x_0) + \frac{d \sin(x_0)}{dx_0} (x - x_0) \dots$$



$$M_d l^2 \ddot{\theta} = -M_g l \sin(\theta) - D \dot{\theta} + \tau$$

$$M_d l^2 \ddot{\theta} + M_g l \sin(\theta) + D \dot{\theta} = \tau \rightarrow M_d l^2 \Delta \ddot{\theta} + D \Delta \dot{\theta} + M_g l (\sin(\theta_0) + \cos(\theta_0) \Delta \theta) = \tau \rightarrow$$

$$\theta = \theta_0 + \Delta \theta \quad \ddot{\theta} = \Delta \ddot{\theta}$$

$$\dot{\theta} = \Delta \dot{\theta}$$

$$M_g l \sin(\theta_0) = \tau_0 \quad \tau = \tau_0 + \Delta \tau$$

$$M_d l^2 \Delta \ddot{\theta} + D \Delta \dot{\theta} + M_g l (\sin(\theta_0) + \cos(\theta_0) \Delta \theta) = \tau_0 + \Delta \tau$$

$$\sin(\theta) = \sin(\theta_0) + \cos(\theta_0) \Delta \theta$$

$$\frac{\Delta \theta}{\Delta \tau} = \frac{1}{M_d l^2 s^2 + D s + M_g l \cos(\theta)} = P(s)$$

